Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How load does it take to download the file?

Reasoning + Answer:

<think> To solve this problem, let's first calculate how much of the file Carla downloaded before the restart. Since the file size is 200 GB and she had 40% of the way through the download before the restart, we can find this by calculating 40% of 200 GB. Let's denote 40% of 200 GB as \(0.40 \times 200 \). So, \(0.40 \times 200 = 80 \) GB. This means she downloaded 80 GB before the restart. Next, we need to account for the 20 minutes it took for the restart to install updates. During these 20 minutes, she didn't make any progress in downloading the file since the download was interrupted. Now, let's focus on the remaining part of the download. Since the total file size is 200 GB and she had already downloaded 80 GB, we can subtract 80 GB from the total file size to find out how much she still needs to download: \(200 - 80 = 120 \) GB. Finally, to find out how long it takes to download the remaining 120 GB at her normal speed of 2 GB per minute, we can divide the remaining GB by her download speed: \(120 \div 2 = 60 \) minutes. Now, we need to add the time she spent downloading the first 80 GB, which is part of the total time, to the 20 minutes for the restart. So, we add 80 minutes (initial download) to 20 minutes (restart time): \(80 + 20 = 100 \) minutes. Thus, the total time to download the file is 100 minutes. </think> <answer> 100 </answer>